

1. Overview

This folder contains the complete "audit trail" for the identification phase of our SLR. To ensure reproducibility, we have documented the exact Boolean strings, database filters, and the raw metadata for the **639 records** initially identified.

2. Detailed Search Strings

The search strategy was executed on March 16, 2026, targeting five major academic databases. The strings were specifically designed to capture the intersection of Insider Threat Detection and Data-Centric Machine Learning (Preprocessing and Feature Engineering).

SEARCH STRINGS AND NUMBER OF RECORDS RETRIEVED.		
Database	Search String	Results
Web of Science	("Artificial Intelligence" OR "Machine Learning" OR "Deep Learning" OR "Neural Networks") AND ("Feature Engineering" OR "Feature Extraction" OR "Feature Selection" OR "Behavior Indicators" OR "Behavioral Features") AND ("Insider Threat Detection" OR "Insider Threat Detection System" OR "Insider Threat" OR "Cyber Threats" OR "Computer Threats" OR "Cybersecurity" OR "Security Threats") AND ("User Behavior" OR "User Behaviour" OR "Behavioral Detection" OR "Behavioral-based Detection" OR "User Behavior-based Detection" OR "Behavioral")	181
Scopus	TITLE-ABS-KEY (("insider threat*" OR "malicious insider") AND ("feature engineer*" OR "feature extract*" OR "feature selection" OR "representation learning" OR "behavior* indicator*" OR "preprocess*" OR "machine learning" OR "deep learning" OR "anomaly detection")) AND PUBYEAR > 2020 AND PUBYEAR < 2026 AND (LIMIT-TO (DOCTYPE , "ar")) AND (LIMIT-TO (LANGUAGE , "English"))	174
IEEE Xplore	("insider threat") AND ("machine learning" OR "deep learning" OR "anomaly detection") AND ("feature engineering" OR "feature extraction" OR "representation learning" OR "preprocessing" OR "behavioral indicators")	116
Science Direct	("insider threat") AND ("feature engineering" OR "representation learning") AND ("machine learning" OR "deep learning") AND ("log" OR "logs")	120
ACM Digital Library	"query": ("insider threat" AND ("machine learning" OR "deep learning" OR "anomaly detection") AND ("feature engineering" OR "feature extraction" OR "representation learning" OR "preprocessing" OR "behavioral indicators")) "filter": Article Type: Research Article, Survey, E-Publication Date: (01/01/2021 TO 12/31/2025)	48
Total Studies Identified		639

2. Files with Raw Results

To allow for independent verification, the table below maps the results to the specific files stored in this directory. All files were exported on March 16, 2026.

Database	Result Count	Primary Export Filename	Format
Web of Science	181	Wos 181 Recods Retrived 16 March 2026	CSV
Scopus	174	scopus_174_records_export_Mar 16	CSV
ScienceDirect	120	Science Direct 120 Records	CSV
IEEE Xplore	116	IEEE 116 Records	CSV
ACM Library	48	acm 48 Records 16 Mar 2026	CSV

3. Multi-Stage Filtration Summary

Following the initial identification of **639** records, we applied a systematic filtration process to arrive at the final **110** primary studies. This process was conducted in accordance with **PRISMA 2020** standards and guidelines.

Database	Stage 1 (Full search Count)		Stage 2 Removing duplicates	Stage 3 (title and Abstract)	Stage 4 (Full Text)	Final Count
	Initial Results	Filter by Year (2021- 2025 Journals Only				
Web of Science	181	118	350	124	117	110
Scopus	174	174				
Science Direct	120	89				
IEEE Explore	116	11				
ACM Digital Library	48	5	350	124	110	110
Remaining Papers	639	397				
Excluded	242		47	226	7	

As illustrated in the filtration table above, the screening process involved the following steps:

- **Stages 1 and 2:** Records outside the 2021–2025 range, non-English papers, and non-journal publications (e.g., conference papers or book chapters) were removed.
- **Stages 3 and 4:** Title and Abstract Screening, followed by a full-text eligibility assessment. Papers were screened for technical depth. Any paper that focused on general ML architectures without specific discussion on data preprocessing or feature engineering for insider threats was excluded.
- **Final Inclusion:** 110 studies provided sufficient technical evidence to answer **RQ1–RQ4**.

The next folder (02_Study_Selection) gives more details on the screening and selection process following the PRISMA methodology.